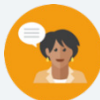


6.4 Connecting One Variable Inequalities to Two-Variable Inequalities

Lesson Introduction

 Benchmarks	MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.		 Learning Target	I can connect the solutions to one-variable inequalities to the solutions of two-variable inequalities using graphs.
	MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set of a two-variable linear inequality.			
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.2.2 Solve two-step, linear inequalities in one variable, and represent solutions algebraically and graphically.		MA.912.AR.3.3 Given a mathematical or real-world context, write and solve one-variable, quadratic inequalities over the real number system. Represent solutions algebraically or graphically. MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable, absolute value inequalities. Represent solutions algebraically or graphically.	
 Essential Question	What is the relationship between the solutions to one-variable inequalities and the solutions of two-variable inequalities?			
 Lesson Narrative	In this lesson, students begin extending their understanding inequalities to two variables. Students will transition from solving one-variable inequalities to graphing and finding solutions of two-variable inequalities on the coordinate plane. They will begin by exploring the relationship between a one-variable inequality and a two-variable inequality, and then transition to exploring linear inequalities in two variables and graphs of their solutions. Students will bring together what they've learned to formalize the learning around the graph of a two-variable inequality.			
 Component Summary	6.4.1 Warm-Up (~5 min)	History of mathematics activity (<i>SE p. 311</i>)	6.4.3 Exploration (15 min)	Explore solutions to two-variable inequalities (<i>SE p. 314</i>)
	6.4.2 Exploration (15 min)	Explore relationship between one-variable and two-variable inequalities (<i>SE p. 312</i>)	6.4.4 Bring It Together (5-10 min)	Summarize conclusions about representing solutions to two-variable inequalities from explorations (<i>SE p. 316</i>)
	6.4.5 Wrap-Up (~5 min)	Coordinate plane reflection activity (<i>SE p. 317</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Warm-Up 6.4.1 contains an optional video to accompany the question prompts.	
	Additional Preparation This lesson is comprised primarily of Exploration activities. Plan your groupings and logistics accordingly to maintain instructional momentum.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse when to use a dotted or solid boundary line when graphing a two-variable inequality. - Students may shade in the incorrect region when graphing a two-variable inequality.	- Pacing in this lesson will be something to keep an eye on. The bulk of the activities are Exploration activities. Consider going through these on your own to look for points of potential misconceptions and determine pacing adjustments that may need to be made. - The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions. There will be time in the remaining lessons in this unit to formalize learning, but these explorations set the foundation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Explorations 6.4.2 and 6.4.3 provide students with collaborative exploration opportunities to leverage the Think-Pair-Share routine. Students will have the opportunity to work on their own and then share their findings and results with a partner for an academic discussion.	MA.K12.MTR.5.1: Patterns and Structure 6.4.2 Exploration has students use what they learned in prior lessons about one-variable inequalities to connect to the concept of a two-variable inequality. During this process students will be relating previously learned concepts to new concepts as they develop a deeper understanding.
 Differentiate Instruction	Consider creating a visual reference for students at the end of this lesson to use as they progress through the remainder of this unit. For example, in 6.4.4 Bring It Together the information could be used on an anchor chart, along with a visual representation of the relationship of the inequality symbol, the open/closed circle, the boundary line, and shading.	
Lesson Notes:		

6.4.1 Warm-Up: History of Mathematics

Component Overview: Students will watch a video about the invention of the coordinate system and answer two questions independently.



6.4.1 Warm-Up: History of Mathematics

René Descartes (1596-1650) was a French philosopher and mathematician. He is credited with the invention of the modern coordinate system, which is also called the Cartesian coordinate system — named after him. As the story goes, Descartes was sick in bed one day and saw a fly crawling on the ceiling. He determined that he could use just two numbers, one for the fly's horizontal position and one for its vertical position, to describe the fly's location at any point.



For this activity, students can watch the accompanying video or read the narrative without watching the video.

1. As the story goes, how did the fly help Descartes invent the coordinate grid system?

Answers will vary. The fly help Descartes invent the coordinate grid system because it provided the problem of how to describe its position at any moment.

2. Name at least one way in which a coordinate system, either in two or three dimensions, is used in everyday life.

Answers will vary. One use of the coordinate system is on a map.

6.4.2 Exploration: Relating One-Variable and Two-Variable Inequalities

Component Overview: Students will explore the relationship between what they have learned about one-variable inequalities and the new concept of two-variable inequalities. For this activity, students will need a partner.



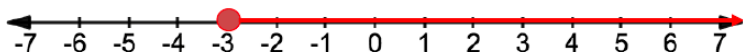
6.4.2 Exploration: Relating One-Variable and Two-Variable Inequalities

1. Use the inequality $-2x + 3 \leq 9$ to complete the following.

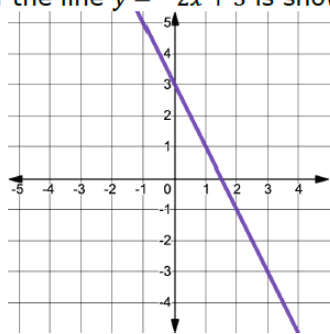
- a. Solve the inequality.

$$\begin{aligned} -2x + 3 &\leq 9 \\ -2x &\leq 6 \\ x &\geq -3 \end{aligned}$$

- b. Graph the solution set.



2. The graph of the line $y = -2x + 3$ is shown.



- a. Use either the graph or the equation to determine the y -values that complete the ordered pairs such that they satisfy the equation $y = -2x + 3$.

$(-2, 7)$	$(0, 3)$	$(-4, 11)$
$(-3.5, 10)$	$(7, -11)$	$(-3, 9)$



This exploration is scaffolded to allow students to engage with the material with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in both this and the next exploration component in the 6.4.4 Bring It Together. Students may revise their thinking throughout the exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Bring It Together.



Question 1 is a refresher of the content learned in the prior lessons. Watch for student performance on this question, as it can provide information on whether students may need additional support with solving and/or graphing the solution to a one-variable inequality.



What method did you use to determine the y -values of the ordered pairs?

6.4.2 Exploration: Relating One-Variable and Two-Variable Inequalities (continued)

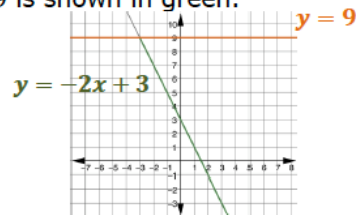
- b. Organize the points in the table.

Ordered Pairs With y-values Less Than or Equal to 9	Ordered Pairs With y-values Greater Than 9
$(0,3)$ $(-2,7)$ $(-3,9)$ $(7,-11)$	$(-4,11)$ $(-3.5,10)$

- c. What do you notice about the x -values of the ordered pairs whose y -values are less than or equal to 9?

Answers will vary. The x -values of the ordered pairs whose y -values are less than or equal to 9 are all greater than the x -values of the ordered pairs whose y -values are greater than 9.

- d. On the coordinate grid, the graphs of $y = -2x + 3$ and $y = 9$ are shown. The part of the graph of $y = -2x + 3$ where the y -values are less than or equal to 9 is shown in green.



For which x -values are the y -values of the equation $y = -2x + 3$ less than or equal to 9?

- ☐ $x \geq -3$
- ☐ $x \leq -3$
- ☐ $x \geq 9$
- ☐ $x \leq 9$

- e. Explain how your answer relates to the solution set of the inequality $-2x + 3 \leq 9$.

Answers will vary. The x -values for which the y -values of the equation $y = -2x + 3$ are less than or equal to 9 are the same as the solution set of $-2x + 3 \leq 9$.



Listen for how students are responding to question 2c when circulating the room. For students that are making generalized statements about things they notice (i.e., there are no repeating x -values, etc.), try to steer them in the direction of the relationship between the x -values to their corresponding y -values so that the connection can be established as the exploration continues.



When circulating during this activity, use what students are saying about question 2e to see what connections, if any, students are making based on the series of scaffolded questions. If they are getting off track, consider providing some prompting questions about things they may have overlooked in the exploration process.

6.4.2 Exploration: Relating One-Variable and Two-Variable Inequalities (continued)

3. Consider the inequality $-2x + 3 \leq -5$.

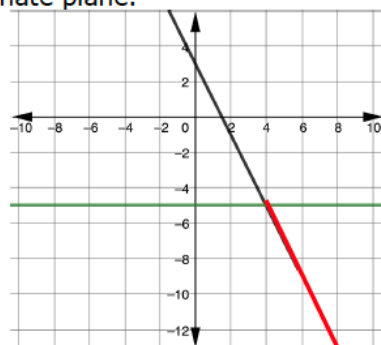
a. Solve the inequality.

$$-2x + 3 \leq -5$$

$$-2x \leq -8$$

$$x \geq 4$$

b. The graphs of $y = -2x + 3$ and $y = -5$ are shown on the coordinate plane.



Highlight the part of the graph of $y = -2x + 3$, where the y -values are less than or equal to -5 .

Shown on coordinate grid.

- c. Describe the relationship between your solution to the inequality in Part A and the values of x for which the graph of $y = -2x + 3$ is less than or equal to -5 .

Answers will vary. In the part of the graph that I highlighted, all the ordered pairs have an x -value that is greater than or equal to 4, which matches the solution set from the inequality in part a.

4. Discuss with your partner how the graph of a line in slope-intercept form, $y = mx + b$, relates to the solution of the related one-variable inequality, $mx + b \leq k$, for some constant k . Summarize your discussion.

Answer will vary. When you solve the one-variable inequality $mx + b \leq k$, the solution set tells you the values of x for which the graph of the line $y = mx + b$ is less than or equal to k .



At the conclusion of the activity, engage students in a whole-group discussion around their responses to question 4. Use this as an opportunity to ensure students have reached the desired outcome of this Exploration activity. Be careful not to give away too much information yet, as they will not tie everything together until 6.4.4 Bring It Together.

6.4.3 Exploration: Linear Inequalities in Two Variables

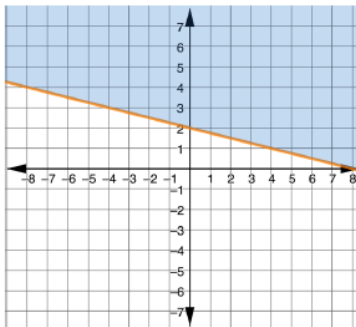
Component Overview: Students will engage in a scaffolded exploration of linear inequalities in two variables using two different inequality problems. For this activity, students will need a partner.



6.4.3 Exploration: Linear Inequalities in Two Variables

The solution sets of one-variable linear inequalities can be represented on a number line. The solution sets of two-variable linear inequalities are represented on the coordinate plane. Work with your partner to explore the graphs of two-variable linear inequalities.

- The graph of the two-variable linear inequality $y \geq -\frac{1}{4}x + 2$ is shown.



- The orange line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

$$y = -\frac{1}{4}x + 2$$

- How does the equation of the boundary line relate to the inequality?
Answers will vary. The equation of the boundary line is the same as the inequality except that it has an equal sign instead of an inequality symbol.
- Select one ordered pair for each of the criteria given in the table, then substitute each ordered pair into the inequality.
Answers will vary.



This lesson has back-to-back Exploration activities. To avoid loss of instructional momentum, consider guiding students through the narrative and creating a generalized discussion using the questions below about the graph to get students started and establish this exploration as a unique activity in the instructional framework.

- What do you notice about this graph?
- How is this graph similar to or different from the graph of a line?
- How is it similar to or different from the graph on an inequality in one variable?

6.4.3 Exploration: Linear Inequalities in Two Variables (continued)

	An Ordered Pair in the Shaded Region	An Ordered Pair in the Unshaded Region	An Ordered Pair on the Boundary Line
Coordinates of Selected Point	(2,2)	(2,0)	(4,1)
Substitute the Point into the Inequality $y \geq -\frac{1}{4}x + 2$	$2 \geq -\frac{1}{4}(2) + 2$ $2 \geq -\frac{1}{2} + 2$ $2 \geq 1\frac{1}{2}$	$0 \geq -\frac{1}{4}(2) + 2$ $0 \geq -\frac{1}{2} + 2$ $0 \geq 1\frac{1}{2}$	$1 \geq -\frac{1}{4}(4) + 2$ $1 \geq -1 + 2$ $1 \geq 1$
Is it a solution to $y \geq -\frac{1}{4}x + 2$?	☒ Yes ☐ No	☐ Yes ☒ No	☒ Yes ☐ No

- d. Describe what part(s) of the graph represents the solution set of the inequality $y \geq -\frac{1}{4}x + 2$.
Answers will vary. The parts of the graph that represent the solution set are the boundary line and the shaded area.

- e. Which part of the graph of the solution set of $y \geq -\frac{1}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y = -\frac{1}{4}x + 2$	☒ boundary line ☐ shaded region
$y > -\frac{1}{4}x + 2$	☐ boundary line ☒ shaded region

- f. Discuss with your partner how you think the graph of $y \leq -\frac{1}{4}x + 2$ would look different from the graph of $y \geq -\frac{1}{4}x + 2$.
Answers will vary. The graph will be shaded below the boundary line rather than above it.



What method did you use to determine that the coordinates were a solution to the inequality?



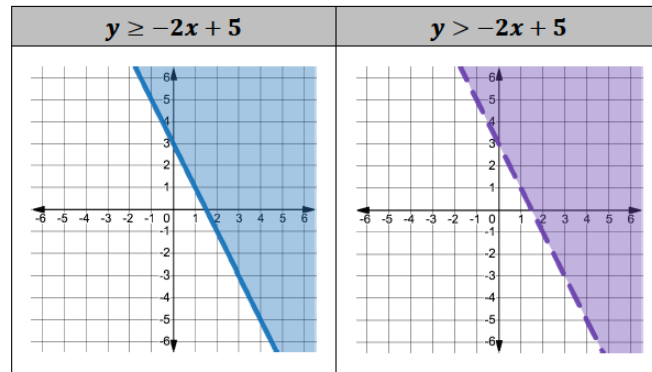
What do you think is the relationship between the boundary line, the shaded region, and a solution to each?



When circulating during this activity, use what students are saying about question 1f as a formative assessment to see what connections, if any, students are making based on the series of scaffolded questions. If they are getting off track, consider providing some prompting questions about things they may have overlooked in the exploration process.

6.4.3 Exploration: Linear Inequalities in Two Variables (continued)

2. Two linear inequalities and their graphs are shown.



- a. Discuss with your partner why you think the boundary line on the graph of $y \geq -2x + 5$ is solid while the boundary line on the graph of $y > -2x + 5$ is dotted. Summarize your discussion in the table.

What do you think a solid boundary line means?	What do you think a dotted boundary line means?
<i>Answers will vary. I think a solid boundary line means that points on the line are in the solution set.</i>	<i>Answers will vary. I think a dotted boundary line means that points on the line are not in the solution set.</i>

- b. Explain how the solid and dotted lines on the graphs of two-variable inequalities relate to closed and open circles on the graphs of one-variable inequalities.
Answers will vary. The solid line is like a closed circle as the solution set includes the boundary. The dotted line is like an open circle, where the solution set does not include the boundary.



- What is the relationship between the type of boundary line and the inequality symbol used?
- Is 3 a solution to both inequalities? Explain how you know.

6.4.4 Bring It Together: Linear Inequalities in Two-Variables

Component Overview: Students will bring together what they have learned in the lesson explorations to formalize understanding of boundary lines when graphing a two-variable inequality on the coordinate plane.



6.4.4 Bring It Together: Linear Inequalities in Two-Variables

The graph of a linear inequality in two variables includes a boundary line and shading on one side or the other side of the boundary line to represent all the ordered pairs that satisfy the inequality.

1. Complete the statements.

If the points on the boundary line are included in the

solution set, the boundary line is ☐ dotted.
☐ solid. If the

points on the boundary line are not included in the

solution set, the boundary line is ☐ dotted.
☐ solid.



On the graph of a linear inequality in two variables, the boundary line is **solid** when the expressions compared are $\geq$ or $\leq$ to each other. If they are $>$ or $<$, the boundary line is **dotted**.

The other ordered pairs that are part of the solution set are represented by the shaded region. Test points on either side of the boundary line can be selected and substituted into the inequality to determine which region should be shaded to represent the solution set.



Consider creating a visual reference for students to use as they progress through the remainder of this unit. For example, this information could be used on an anchor chart, along with a visual representation of the relationship of the inequality symbol, the boundary line, and shading.

6.4.5 Wrap-Up: Reflection

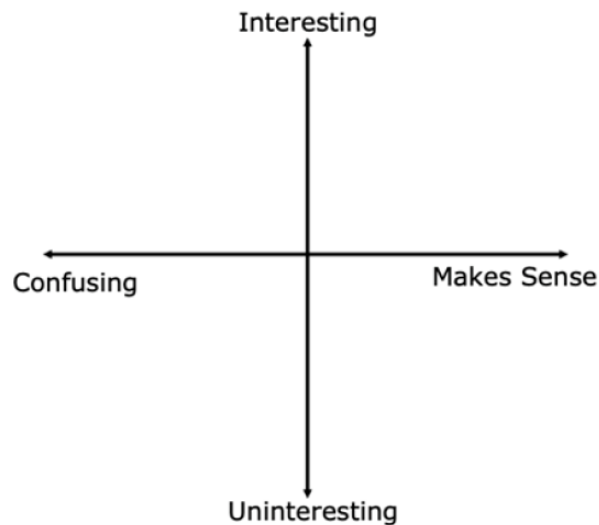
Component Overview: Students will complete a self-reflection where they will plot the intersection of their level of interest, and how much the content of this lesson made sense to them.



6.4.5 Wrap-Up: Reflection

- Plot a point on the grid below to indicate your level of interest and how much sense you made of today's learning target.

Answers will vary.



- Explain why you chose that point and that quadrant.

Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.